MLS Goalkeeper & Match Analysis

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Agenda

Topics covered in this presentation

DATA & ANALYSIS

MODELING & STATISTICS

Overview

Datasets, tools, and analysis components

DATASETS

- **all_goalkeepers.csv**
  Season-level GK stats (2012–2020)
- **matches.csv**
  Match-level results (2012–2020)

AUDIENCE & KEY QUESTIONS

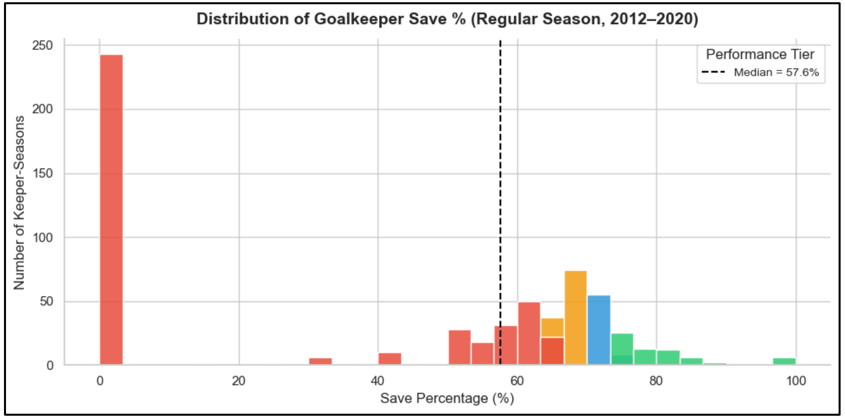
- Soccer Coaches, GMs, and analytical staff

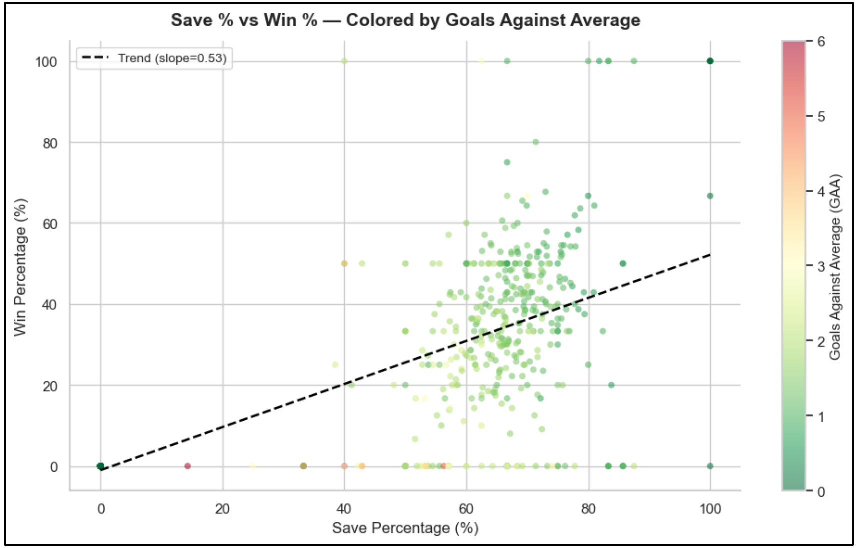
Key Prediction:
Predict winning record from Save %, GAA, Shutouts, Games Played, and Win %

2012–2020	25	2	2
Time Period	Seasons	Models	Datasets

Exploratory Data Analysis

Save % tier classification and analysis





KEY FINDINGS

- Elite tiers (≥75%) strongly correlate with higher win %

Good: 70–74% | Average: 65–69% | Below Average: <65%

- Clear separation between performance groups in win outcomes
- Cross-tabulation confirms Elite tier GKs have significantly higher win %. Save % is a meaningful predictor of team success.

Descriptive Statistics

Key metrics and data quality assessment

METRIC	DETAILS
Key GK Stats	GP (Games Played), minutes, shots faced, saves, GA (Goals Against), GAA (Goals Against Average), save %, win %, shutouts
Save % Median	**≈57.6%**
GAA Median	**≈1.09**

MISSING DATA

- Matches: attendence
- Goalkeepers: Club
- **Overall data quality is high**

57.6%

Save % Median

1.09

GAA Median

Insights (1)

Club performance and home advantage analysis

DEFENSIVE STRENGTH

Clubs with highest career save % historically:
Seattle Sounders, D.C. United, Minnesota United
Rankings vary by era

HOME ADVANTAGE

- Home win % ~52 – 63.3%
- Away win % ~21.4 – 35.3 %
- Strong, persistent home-field effect

~56%

Avg Home Win Rate

~26%

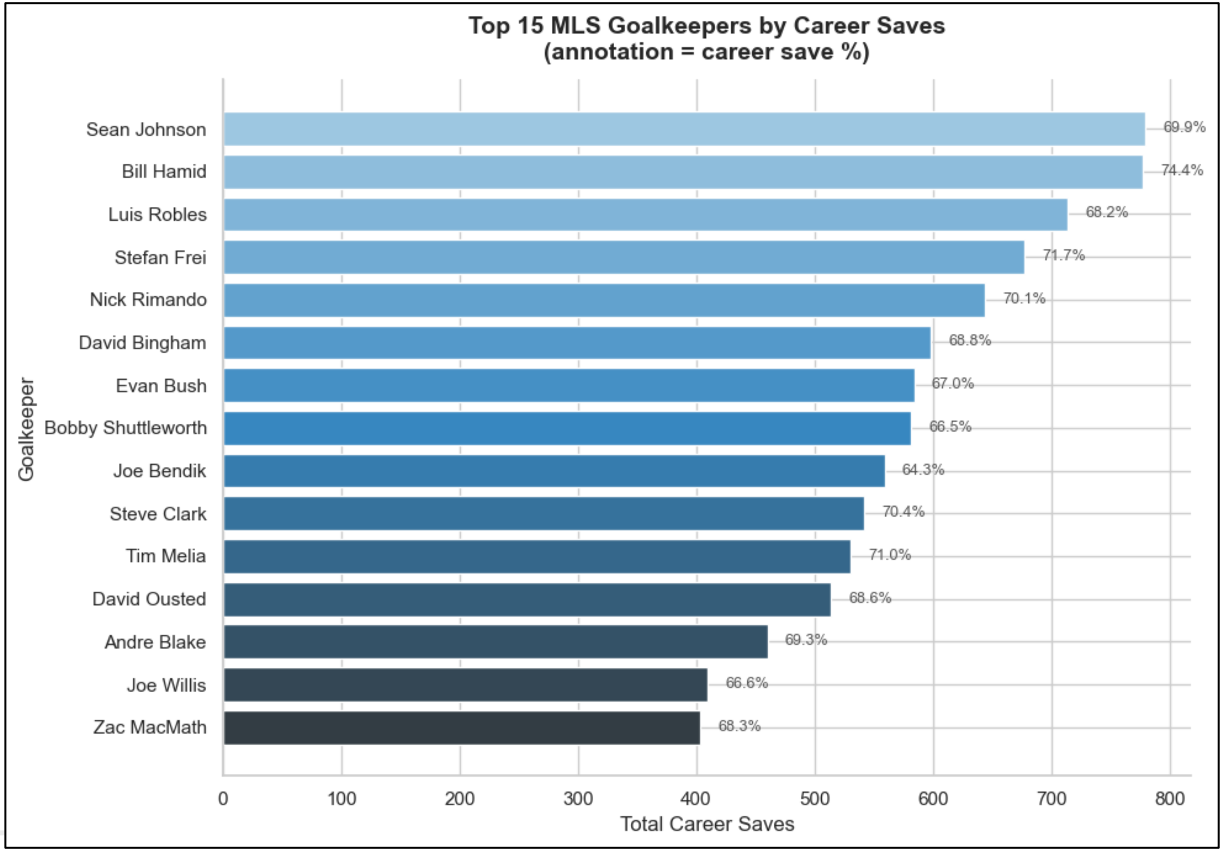
Avg Away Win Rate

Insights (2)

Volume + efficiency view

- High total saves = longevity + workload
  - Played many seasons,
  - Logged high minutes,
  - Played behind a defense that allowed many shots.

- Save% annotation adds efficiency context
  - High-volume, high-efficiency keepers **(elite careers)**
  - High-volume, average-efficiency keepers **(workhorses)**
  - Lower-volume but extremely efficient keepers (not shown because they didn't make top 15 in saves)



CHICAGO BOOTH

Predictive Modeling

Logistic Regression — Predicting Winning Records

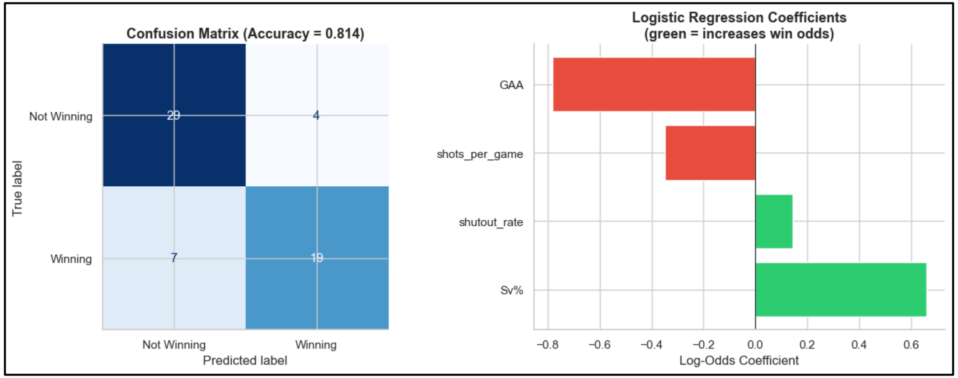
GOAL: Predict winning record (Win > Loss seasons) using GK performance metrics

0.81

Model Accuracy (81%)

TOP PREDICTORS

Save % — positive influence on winning | **GAA** — strong negative (higher GAA hurts) | **Shutout Rate** — slight negative effect



Key Takeaways & Predictions

Major findings from the full Colab analysis

LEAGUE TRENDS

- Save % remains stable across eras
- GAA has improved over time
- Home advantage is strong and consistent (~56% home win rate)

KEEPER INSIGHTS

- Rimando, Hartman, and Meola dominate career metrics
- Elite save % tiers (≥75%) strongly correlate with winning seasons

MODELING

- Logistic regression achieves 81% accuracy predicting winning seasons
- GAA is the strongest negative predictor; Save % is the strongest positive predictor

BOTTOM LINE: Goalkeeper save % and goals-against average are reliable, actionable metrics for predicting MLS team success – Get a Great Goalie!!